Requirements Document

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Version and Change Log

Version	Date	Change Description	Author
0.1	2026-09-16	Initial project constraints (Economic, Professional, Legal, Security) and team contract baseline established in Assignment 3.	Matthew Goldsberry, Caden O'Leary, Elmoataz Abdellah, Seunghun Lee, Kristen Griessman
0.2	2026-09-28	Added Week 4 stakeholders, user stories (US-01 through US-04), use case UC-01, acceptance criteria (AC-01.1 through AC-01.3), and initial L4 OBC requirements draft.	Caden O'Leary, Matthew Goldsberry, Elmoataz Abdellah
0.3	2026-10-14	Expanded L3 (L3-OBC-129 to L3-OBC-176) and L4 (L4-OBC-185 to L4-OBC-268) OBC requirements for D1/D2 subsystem interfaces, ASCON-AEAD128 command authentication, Data Processor offloading, and watchdog fault recovery. Withdrew draft requirements No. 177-184 (L3-OBC-177 through L3-OBC-184) without renumbering to preserve stable IDs.	Seunghun Lee, Kristen Griessman, Matthew Goldsberry
1.0	2026-10-28	Finalized Assignment 9 Requirements Document using the L0-L4 requirements baseline from the Assignment 9 tab. Documented formal opt-out of the MoSCoW summary in favor of CubeCats modal-verb requirement formatting ("shall" / "should"), separated Functional and Non-Functional L3-OBC/L4-OBC requirement tables, mapped Week 3 constraints (CON-01 to CON-10), and logged assumptions and open questions.	Caden O'Leary, Elmoataz Abdellah, Matthew Goldsberry, Seunghun Lee, Kristen Griessman

Purpose and Scope

Purpose

In collaboration with the University of Cincinnati CubeCats organization, this Requirements Document establishes the functional, non-functional, interface, and regulatory baseline for the initial prototype flight software of the On-Board Computer (OBC) on the HABSat-1 3U CubeSat mission. HABSat-1 is designed to capture multispectral imagery (620 nm, 650 nm, 680 nm, and 708 nm wavelengths with a 10 nm FWHM bandwidth) and truth imagery of Harmful Algal Blooms (HABs) in Lake Erie from Low Earth Orbit (LEO) at a Ground Sampling Distance (GSD) below 90 meters per pixel (L0-PRJ-001, L1-SCI-008, L1-SCI-009). This document governs flight software implementation, software-in-the-loop (SITL) and hardware-in-the-loop (HITL) verification, and Preliminary Design Review (PDR) evaluation.

Scope

The HABSat-1 OBC flight software executes on the primary Main Computer (EnduroSat OBC Type I board with an STM32F407VG ARM Cortex-M4 microcontroller running FreeRTOS and the NASA F' framework) and coordinates with the dedicated secondary Data Processor (RP2350 coprocessor). Within scope for the OBC flight software are:

- **Command and Data Handling (CDH):** Receiving, decrypting (L3-OBC-129, L4-OBC-185), authenticating (L3-OBC-130, L4-OBC-186–189), validating (L3-OBC-131, L4-OBC-190–193), queuing (L3-OBC-168, L4-OBC-197–198), and dispatching uplinked telecommands from the Ground Station, and returning positive/negative acknowledgments (L3-OBC-132, L4-OBC-194–196).
- **Autonomous Mission & Pass Sequencing:** Managing the deterministic 10-mode Finite State Machine across Pre-Launch, Initialization, Checkout, Standby, Imaging, Downlink, Repeater, Low-Power, Safe, and Emergency modes (L3-OBC-137, L4-OBC-209–211) and executing the event-driven orbital imaging pass sequencer (L3-OBC-151–154, L4-OBC-234–243).
- **Subsystem Orchestration & Health Monitoring:** Polling telemetry from all onboard subsystems every 60 seconds (L3-OBC-138, L4-OBC-220–221, L4-OBC-231–233) and handling asynchronous fault alerts within 100 ms (L3-OBC-139, L4-OBC-222) across the ADCS, Power Computer (EPS), Imager Adapter, Data Processor, Data Storage, UHF Telemetry/Command Radio, S-Band Main Downlink Radio, Secondary Payload, and Deployment Mechanisms.
- **On-Board Science Processing & Storage Management:** Commanding Chloropyhll Index (CI) calculation on the Data Processor (L3-OBC-167, L4-OBC-244–246), tagging images with synchronized ECEF position and LVLH attitude metadata (L3-OBC-173, L4-OBC-260), performing lossless compression and CRC verification (L3-OBC-146, L4-OBC-223–224), and retaining captured images in non-volatile memory until an explicit Ground Station purge command is received (L3-OBC-147, L4-OBC-225).
- **Fault Detection, Isolation, and Recovery (FDIR):** Managing hardware watchdog timers (L4-OBC-250, L4-OBC-258), internal bus fault recovery and node isolation (L3-OBC-160, L4-OBC-252, L4-OBC-263), golden-image boot recovery (L3-OBC-166, L4-OBC-251), and autonomous Safe Mode transitions (L3-OBC-161–162, L4-OBC-254–255).

Stakeholders and User Goals from Week 4

Stakeholder Classification

Stakeholder	Category	Primary Role & Interest in HABSat-1 OBC
Satellite Operator	Primary	Commands spacecraft operations from the Ground Station, monitors subsystem health telemetry, uploads TLEs and target coordinates, and verifies command execution.
Environmental Data Researcher	Secondary	Relies on radiometrically accurate, georeferenced multispectral imagery and Chloropyhll Index maps downlinked from HABSat-1 to analyze harmful algal blooms in the Great Lakes.
Future Club Members	Secondary	Inherits the HABSat-1 flight software repository, NASA F' architecture, hardware-mocked unit test suite, and onboarding documentation for subsequent CubeCats satellite missions.
Amateur Radio Operator	Hidden	Receives unencrypted 49-second UHF housekeeping beacons and utilizes the spacecraft's UHF amateur radio repeater in accordance with amateur radio regulations.

User Stories (Week 4 Baseline)

- **US-01 (Primary dSatellite Operator):** As a satellite operator, I want to send authorized commands to HABSat-1 and receive confirmation that the commands were processed, so that I can control spacecraft operations and verify that the satellite received my instructions.
- **US-02 (Secondary - Environmental Data Researcher):** As an environmental data researcher, I want HABSat-1 to collect multispectral images with their corresponding observation time and mission data, so that I can use the collected data to analyze harmful algae blooms in the Great Lakes.
- **US-03 (Secondary - Future Club Members):** As a future CubeCats member working on the next satellite mission, I want to be able to learn from the HABSat-1 flight software documentation and development decisions, so that I can understand how the previous mission was designed and use that knowledge when developing the next satellite.
- **US-04 (Hidden - Amateur Radio Operator):** As a licensed amateur radio operator communicating with HABSat-1, I want to understand and follow the satellite's communication requirements and procedures, so that I can communicate with the satellite in accordance with the mission's operating procedures and applicable amateur radio regulations.

Core Use Case & Acceptance Criteria (UC-01: Send and Process Spacecraft Command)

- **Story:** US-01 | **Primary Actor:** Satellite Operator | **Secondary Actors:** Ground Station, HABSat-1 On-Board Computer (OBC), Satellite Subsystem
- **Preconditions:**
    1. HABSat-1's OBC is powered on and the flight software is running.
    2. The ground station has an established communication link with HABSat-1.
    3. The satellite operator is authorized to send commands to HABSat-1.
    4. The command is supported by the flight software.
    5. The command is valid for the satellite's current mission flight state.
- **Main Success Flow:**
    1. **Satellite Operator:** Selects an authorized command to send to HABSat-1.
    2. **Ground Station:** Formats and transmits the command to HABSat-1 using the established communication protocol.
    3. **OBC:** Receives the command from the ground station.
    4. **OBC:** Validates the command's format, authorization, and compatibility with the current mission flight state.
    5. **OBC:** Routes the validated command to the appropriate satellite subsystem.
    6. **Satellite Subsystem:** Receives and executes the command.
    7. **Satellite Subsystem:** Returns the execution status to the OBC.
    8. **OBC:** Records the command and its execution status.
    9. **OBC:** Transmits a command confirmation to the ground station.
    10. **Ground Station:** Receives the confirmation and makes the command status available to the satellite operator.
    11. **Satellite Operator:** Verifies that the command was successfully processed.
- **Alternate Flow (A1: Command is scheduled for later execution):**
    1. **Satellite Operator:** Sends a valid command with a specified future execution time.
    2. **OBC:** Validates the command and verifies that the scheduled execution time is valid.
    3. **OBC:** Stores the command in the scheduled task queue.
    4. **OBC:** Sends confirmation to the ground station that the command has been accepted for scheduled execution.
    5. **OBC:** Waits until the scheduled execution time.
    6. **OBC:** Executes the command and sends the execution status to the ground station.
- **Exception Flow (E1: Invalid or unauthorized command):**
    1. **Satellite Operator:** Sends a command to HABSat-1.
    2. **OBC:** Receives the command and determines that the command is malformed, unauthorized, unsupported, or invalid for the current mission flight state.
    3. **OBC:** Rejects the command without sending it to the affected subsystem.
    4. **OBC:** Sends a command-rejection status to the ground station.
    5. **Ground Station:** Presents the rejection status to the satellite operator.

- **Postconditions:** If the command is valid and successfully executed, the requested spacecraft operation has been performed and the command's execution status has been recorded and reported to the ground station. If the command is rejected, the requested operation has not been performed and the rejection and reason have been reported to the ground station.
- **Acceptance Criteria:**
    - **AC-01.1 (Successful Command):** Given the OBC is running, the ground station has an established communication link, and the satellite operator is authorized to send a command that is valid for the current mission flight state, when the satellite operator sends the command to HABSat-1, then the OBC shall validate the command, route it to the appropriate subsystem, and provide the ground station with the command's execution status.
    - **AC-01.2 (Scheduled Command):** Given the OBC is running and receives a valid command with a valid future execution time, when the scheduled execution time is reached, then the OBC shall execute the command and record the resulting execution status.
    - **AC-01.3 (Invalid Command):** Given the OBC receives a command that is malformed, unauthorized, unsupported, or invalid for the current mission flight state, when the OBC validates the command, then the OBC shall reject the command, shall not send it to the affected subsystem, and shall record the reason for rejection.

Glossary of Domain Terms and Role Names

Term / Role Name	Definition
ADCS (Attitude Determination and Control System)	The spacecraft subsystem (attitude sensors, reaction wheels, magnetorquers, GPS receiver, and ADCS Computer) responsible for determining orbital position/orientation and pointing the satellite.
Amateur Radio Operator	A hidden external stakeholder holding an amateur radio license who receives public UHF telemetry beacons and relays packets through the HABSat-1 digipeater (US-04).
Analysis	Use of mathematical modeling and analytical techniques to predict the compliance of a design to its requirements using derived or calculated data.
ASCON-AEAD128	The NIST-standardized lightweight Authenticated Encryption with Associated Data (AEAD) cipher (128-bit key, 16-byte IV, 16-byte authentication tag) used to encrypt and authenticate non-amateur HABSat-1 telecommands and private telemetry.
CAN Bus (Controller Area Network)	A noise-immune, differential multi-drop serial bus (CAN 2.0B) using fixed 12-byte frames for deterministic command and telemetry exchange between the Main Computer, ADCS, Power Computer, Imager Adapter, and Data Processor.
CDH (Command and Data Handling)	The core flight software discipline responsible for telecommand ingestion, validation, task scheduling, health monitoring, and telemetry/payload data routing.
CI (Chloropyhll Index)	The NOAA-derived optical band-ratio algorithm applied to 620 nm, 650 nm, 680 nm, and 708 nm multispectral bands to quantify harmful algal bloom concentration (L1-SCI-008, L4-OBC-244).

Term / Role Name	Definition
CRC (Cyclic Redundancy Check)	An error-detecting code appended to communication frames and stored files to detect bit-flip corruption across noisy RF links, serial buses, or flash memory (L4-OBC-224).
CubeCats	The University of Cincinnati student spacecraft engineering organization developing the HABSat-1 mission.
CubeSat (3U)	A nanosatellite conforming to the CubeSat Design Specification with a 3-unit form factor (L2-SYS-026) and a maximum mass of 6,000 grams (L2-SYS-027).
Data Processor (DP)	The dedicated secondary coprocessor (RP2350) that ingests raw frames from the Imager Adapter over UART, computes the Chloropyhll Index (L3-OBC-167), performs lossless compression, and streams imagery to the S-Band radio.
Data Storage (DS)	The onboard non-volatile memory subsystem comprises radiation-tolerant SPI MRAM (for critical state flags and timers, L3-OBC-145) and SPI NAND Flash (for telemetry logs and captured images, L3-OBC-144).
Demonstration	Use of a realized end product to show that a set of stakeholder expectations can be achieved.
Digipeater	A digital amateur radio repeater mode in which the UHF transceiver receives, buffers (L4-OBC-207), and rebroadcasts unencrypted amateur radio packets (L2-MIS-080, L3-OBC-155).
DMA (Direct Memory Access)	Hardware controller capability allowing peripherals (UART, SPI, CAN) to transfer data directly to/from SRAM without saturating the CPU core.
ECEF (Earth-Centered, Earth-Fixed)	A Cartesian spatial reference frame rotating with the Earth, used to tag the geographical position of the spacecraft during image capture (L2-SYS-041, L3-OBC-173).
EnduroSat OBC Type I	The space-qualified PC/104 On-Board Computer hardware platform hosting the primary STM32F407VG ARM Cortex-M4 processor and auxiliary inertial/magnetic sensors.
Environmental Data Researcher	A secondary stakeholder who uses HABSat-1's downlinked multispectral imagery and CI maps to monitor and model harmful algal blooms (US-02).
EPS / Power Computer	The Electrical Power Subsystem controller that regulates solar array MPPT charging, monitors battery State of Charge (SoC), switches subsystem power rails (L4-OBC-210), and monitors the OBC heartbeat (L4-OBC-258).
F' (NASA F Prime)	An open-source (Apache 2.0), component-driven flight software framework developed by NASA JPL featuring static memory allocation and strongly typed message-passing ports.
Faculty Advisor	Dr. Will Hawkins III, UC Computer Science faculty member providing technical oversight, architecture review, and tie-breaking governance for the OBC Senior Design team.
FreeRTOS	An open-source (MIT) preemptive real-time operating system kernel providing deterministic priority-based task scheduling (L4-OBC-247) and mutex/queue primitives on the ARM Cortex-M4.
Future Club Members	Secondary stakeholders consist of incoming UC CubeCats students who will maintain, test, and adapt the HABSat-1 codebase for subsequent missions (US-03).
FWHM (Full Width at Half Maximum)	The spectral bandwidth (10 nm for HABSat-1, L1-SCI-008) across which an optical bandpass filter transmits at least 50% of peak intensity.
GPIO (General-Purpose Input/Output)	Discrete digital hardware pins controlled by the Main Computer to trigger burn-wire deployment mechanisms and read microswitch feedback (L4-OBC-214, L4-OBC-215).

Term / Role Name	Definition
Ground Station (GS)	The University of Cincinnati earth terminal (UHF and S-Band antennas, transceivers, F' GDS, SQLite database, and React dashboard) used to command and monitor HABSat-1.
GSD (Ground Sampling Distance)	The physical distance on the Earth's surface is represented by a single image pixel (required to be <= 90 meters per pixel for HABSat-1, L1-SCI-009, L2-MIS-074, L4-OBC-237).
HAB (Harmful Algal Bloom)	Rapid proliferation of toxic chloropyhll in aquatic ecosystems (such as Lake Erie) targeted by HABSat-1's remote sensing payload (L0-PRJ-001).
I2C (Inter-Integrated Circuit)	A two-wire synchronous serial bus used internally by the ADCS and board-level sensors (L4-OBC-259) for short-distance peripheral communication.
Imager Adapter	The payload interface module that orchestrates the Truth Imager and Multispectral Imager over Camera Link and streams raw image blocks to the Data Processor over UART.
Inspection	Visual examination of a realized end product to validate physical design features or specific manufacturer identification.
ITAR / EAR	U.S. International Traffic in Arms Regulations and Export Administration Regulations governing export control and research security (L1-PRJ-021, CON-06).
LEO (Low Earth Orbit)	An Earth-centered orbit (typically 350–550 km altitude) where HABSat-1 operates (L0-PRJ-001).
LVLH (Local Vertical, Local Horizontal)	An orbital attitude reference frame aligned with the spacecraft's nadir vector and orbital velocity vector, used for camera pointing and georeferencing (L2-SYS-040, L3-OBC-173).
Main Computer (OBC)	The primary flight computer on the EnduroSat OBC Type I board that executes the central state machine, command dispatcher, pass sequencer, and health polling loops.
MPPT (Maximum Power Point Tracking)	The solar charge controller algorithm on the Power Subsystem that dynamically adjusts electrical load impedance to maximize solar energy harvest.
MRAM (Magnetoresistive Random-Access Memory)	Radiation-tolerant non-volatile memory connected via SPI used to store critical boot flags, inhibit timers, and parameter tables (L3-OBC-145, L4-OBC-212, L4-OBC-264).
Multispectral Imager	The primary optical payload camera captures narrowband imagery at 620 nm, 650 nm, 680 nm, and 708 nm to detect harmful algal blooms (L1-SCI-008).
NVM (Non-Volatile Memory)	Persistent storage media (MRAM, NOR Boot Flash, NAND Flash) that retain data across power cycles and processor resets.
OBC Developer / Architect	Team role (Seunghun Lee) responsible for technical direction, hardware-software interface architecture, and storage/coprocessor design.
OBC Documentation Owner	Team role (Caden O'Leary) responsible for maintaining design reports, requirement consistency, and submission formatting.
OBC QA / Test Owner	Team role (Elmoataz Abdellah) responsible for unit/integration test planning, SITL simulation harnesses, and requirement verification.
OBC Security Owner	Team role (Kristen Griessman) responsible for telecommand encryption (ASCON-AEAD128), authentication, key isolation, and security verification.
OBC Team Lead (Coordinator)	Team role (Matthew Goldsberry) responsible for coordinating the Senior Design and club OBC teams, schedules, and subsystem integration milestones.
PC/104	The standardized CubeSat stackable circuit board form factor housing the OBC within its 95 mm x 95 mm x 25 mm volume (L3-OBC-165).

Term / Role Name	Definition
PDR (Preliminary Design Review)	Formal technical milestone review conducted with aerospace industry professionals (CON-03).
PPS (Pulse Per Second)	A hardware timing pulse output by the GPS receiver to synchronize the onboard spacecraft epoch clock (L4-OBC-218).
Satellite Operator	The primary stakeholder role representing authorized UC CubeCats ground personnel who transmit encrypted telecommands and manage orbital passes (US-01).
SEU / SEL (Single-Event Upset / Latch-up)	Radiation-induced transient bit flips (SEU) or parasitic high-current states (SEL) mitigated by shielding, watchdogs, and golden-image boot recovery (L2-SYS-059, L4-OBC-251).
SoC (State of Charge)	The remaining usable electrical energy in the spacecraft battery is expressed as a percentage of full capacity (L4-OBC-203, L4-OBC-237).
SPI (Serial Peripheral Interface)	A 4-wire synchronous serial bus (SCLK, MOSI, MISO, SS) used for high-speed communication between the Main Computer, Data Processor, Data Storage, and S-Band radio.
Test	Use of a realized end product to obtain detailed data to validate performance or to provide sufficient information to validate performance through further analysis.
TLE (Two-Line Element Set)	A standardized orbital ephemeris format uplinked from the Ground Station (0x12) and forwarded to the ADCS (L3-OBC-151, L4-OBC-234) to compute imaging pass schedules.
Truth Imager	The RGB context camera paired with the Multispectral Imager to capture visual reference images of Lake Erie (L4-OBC-223).
UART (Universal Asynchronous Receiver-Transmitter)	An asynchronous serial protocol used for variable-length framed data exchange with the UHF Telemetry/Command Radio, Secondary Payload, and Imager-to-DP stream.
VSWR (Voltage Standing Wave Ratio)	A measure of radio-frequency impedance matching at the antenna feed (EC-41).
WDT (Watchdog Timer)	Hardware and software countdown timers that trigger a processor reboot or Safe Mode transition if not deterministically strobed within their timeout window (L4-OBC-250, L4-OBC-258).

Functional and Non-Functional Requirements

Requirements Language & MoSCoW Opt-Out Acknowledgment

The HABSat-1 project defines requirements using standard aerospace modal verbs to denote requirement criticality, **formally opting out of the MoSCoW prioritization framework** in favor of our hierarchical requirement formatting:

- **"... shall ..." (Mandatory Requirement):** This requirement must be met by the system or project, or the system will be considered to have failed.
- **"... should ..." (Recommended Requirement):** It is strongly recommended to comply with this requirement, but the project will not fail if it is not met.
- **Stable ID & Withdrawal Policy:** Every requirement carries a permanent identifier (No. and L#-XXX-###). IDs are stable and never reused: when draft requirements No. 177 through 184 (L3-OBC-177 to L3-OBC-184) were withdrawn during the D1/D2 transition to the dedicated Data Processor architecture, their numbers were permanently retired rather than renumbering L4-OBC-185 through L4-OBC-268.

Requirement Levels

Level	Type	Definition / Scope
L0	Mission Objectives	What are we trying to accomplish with the project?
L1	Mission Requirements	What do we need to do to accomplish the mission objectives? What do our stakeholders require of the mission?
L2	System Requirements	What does each system as a whole need to do to satisfy mission requirements?
L3	Subsystem Requirements	What does each subsystem need to do to meet the system requirements?
L4	Component Requirements	What special requirements (if any) do hardware/software/firmware components have to accomplish system/subsystem requirements?

Verification Methods

Method	Definition
Analysis	Use of mathematical modeling and analytical techniques to predict the compliance of a design to its requirements using derived or calculated data.
Demonstration	Use of a realized end product to show that a set of stakeholder expectations can be achieved.
Inspection	Visual examination of a realized end product to validate physical design features or specific manufacturer identification.
Test	Use of a realized end product to obtain detailed data to validate performance or to provide sufficient information to validate performance through further analysis.

Requirements

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
1	PRJ	L0-PRJ-001	Science Objective	The HABSat-1 Project shall demonstrate the collection of multispectral images of the Harmful Algal Bloom (HAB) dynamics on Lake Erie from LEO at an increased temporal and spatial resolution compared to current satellite capabilities.	Data collected by HABSat-1 will facilitate analysis of the mechanisms driving changes in these locations and support the development of improved predictive models for HABs in the targeted areas.	Demonstration	N/A
2	PRJ	L0-PRJ-002	Technology Objective	The HABSat-1 Project shall demonstrate the use of a low-cost, drone-rated multispectral imager for remote sensing of natural phenomena from space.	NOAA, DoD, and NASA desire a proof-of-concept of an inexpensive water quality sensing satellite.	Demonstration	N/A
4	PRJ	L0-PRJ-004	External Stakeholder Requirements	The HABSat-1 Project shall comply with all external stakeholder requirements.	Assists in mission success, assures safety, and guarantees legal or regulatory compliance.	Inspection	N/A
8	SCI	L1-SCI-008	Multispectral Imaging Data	Multispectral images collected by the HABSat-1 Project shall cover the 620 nm, 650 nm, 680 nm, and 708 nm wavelengths of light, with a FWHM bandwidth of 10 nm for each wavelength.	The selected wavelengths/bandwidth allow the data to be processed using the NOAA Chloropyhll Index (CI) algorithm.	Demonstration	L0-PRJ-001
9	SCI	L1-SCI-009	Imaging Ground Resolution	Multispectral images collected by the HABSat-1 Project shall have a Ground Sampling Distance (GSD) of no more than 90 meters per pixel.	A target GSD of 70-90 meters is derived from field surveys and multispectral satellite data.	Analysis	L0-PRJ-001
10	SCI	L1-SCI-010	Imaging Georeference	Each multispectral image collected by HABSat-1 Project shall have a coordinate reference locating its geographical center.	The location of any particular image will be required for co-registration of data.	Test, Analysis	L0-PRJ-001
12	SCI	L1-SCI-012	Mission Length	The HABSat-1 project shall collect multispectral imagery of Lake Erie over the course of at least one year.	An adequate amount of data can be collected over the course of one full cycle of seasons.	Analysis	L0-PRJ-001
15	MIS	L1-MIS-015	Flight Element	The HABSat-1 Project shall use a single prototype CubeSat to collect and downlink all multispectral imagery from orbit.	Serves as an explicit statement of the flight element mission and properly scopes this mission.	Inspection	L0-PRJ-002
18	PRJ	L1-PRJ-018	Amateur Radio Support	The HABSat-1 Project should support the amateur radio community by providing communication between	Amateur licensing is one of the simplest paths for radio licensing.	Inspection	L0-EDU-007

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
				operators, educational opportunities, or some other contribution to the radio arts.			
19	PRJ	L1-PRJ-019	Communication Regulation Compliance	The HABSat-1 Project shall comply with all US regulations and procedures for the purposes of licensing and utilization of radio frequencies.	Must comply with FCC regulatory constraints in order to operate radio links.	Inspection	L0-PRJ-004
20	PRJ	L1-PRJ-020	Remote Sensing Regulation Compliance	The HABSat-1 Project shall comply with all US regulations and procedures relating to remote sensing.	Must comply with NOAA regulations on remote sensing from space.	Inspection	L0-PRJ-004
21	PRJ	L1-PRJ-021	Export Control Compliance	The HABSat-1 Project shall comply with all US regulations and procedures relating to export control and research security.	Must comply with the EAR and ITAR regimes.	Inspection	L0-PRJ-004
26	SYS	L2-SYS-026	Form Factor	The CubeSat shall have a 3U form-factor.	Minimum size able to accommodate all required elements with margin.	Inspection	L1-MIS-015
27	SYS	L2-SYS-027	Total Mass	The CubeSat shall have a mass of no more than 6000 grams.	Maximum mass of a 3U Satellite from CubeSat Design Specification.	Inspection	L1-PRJ-023
38	SYS	L2-SYS-038	Imaging Pointing	The CubeSat shall actively point the multispectral imager to capture an entire targeted area during an imaging pass.	Active control is required to point the camera at targets during imaging passes.	Test	L1-MIS-015
40	SYS	L2-SYS-040	Attitude Determination	The CubeSat shall be able to determine its orientation in the LVLH frame during the capture of each image.	Accurate orbital position and attitude required to provide a georeference tag.	Analysis, Test	L1-SCI-010
41	SYS	L2-SYS-041	Position Determination	The CubeSat shall be able to determine its position in the ECEF frame during the capture of each image.	Accurate orbital position and attitude required to provide a georeference tag.	Analysis, Test	L1-SCI-010
42	SYS	L2-SYS-042	Image Data Storage	The CubeSat shall be capable of storing at least 7 multispectral images onboard the satellite at a given time.	Provides buffer time between imaging passes to downlink imagery without build-up.	Inspection	L1-SCI-014
43	SYS	L2-SYS-043	Telemetry Collection	The CubeSat shall be able to collect health/status information of onboard systems for later downlink.	Helps system operators ensure functioning properly and assist with fault diagnosis.	Demonstration	L1-MIS-015
44	SYS	L2-SYS-044	System Lifetime	The CubeSat shall be designed to operate for at least 2 years.	Allows the satellite to complete main scientific mission twice with buffer.	Analysis, Inspection	L1-SCI-012

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
46	SYS	L2-SYS-046	Energy Balance	The CubeSat should maintain a positive orbit-average energy balance in all nominal operational modes.	Consumption should not outpace production over average orbital periods.	Analysis	L1-SCI-014
48	SYS	L2-SYS-048	Safe Mode	The CubeSat shall include a Safe Mode that keeps the satellite power positive, thermally benign, and commandable if unresolvable faults occur.	Helps satellite survive faults and facilitates ground operators during fault diagnosis.	Analysis, Test	L1-SCI-012
49	SYS	L2-SYS-049	Commandability	The CubeSat shall respond to and execute all valid ground station commands it receives.	Commands required to perform basic mission tasks must not cause loss or harm.	Demonstration, Test	L1-MIS-015
50	SYS	L2-SYS-050	Internal Temperature	All internal elements of the CubeSat shall be operational over a temperature range of -40°C to 60°C.	Expected temperature range that internal components may experience on orbit.	Analysis, Inspection, Test	L1-MIS-017
53	SYS	L2-SYS-053	Testing Interfaces	The CubeSat should include communication and power interfaces that facilitate external testing at the subsystem level when the satellite is in an integrated state.	Facilitates testing after final integration when satellite cannot be taken apart.	Inspection	L1-MIS-015
59	SYS	L2-SYS-059	Radiation Survivability	The CubeSat's subsystems should be designed to mitigate the effects of radiation encountered in the mission orbit.	Passage through Van Allen belts and radiation from solar storms can be expected.	Analysis	L1-MIS-016, L1-MIS-017
60	SYS	L2-SYS-060	Heartbeat	The CubeSat should transmit a periodic beacon containing basic health telemetry data when not actively downlinking.	Proves satellite is active/healthy and helps gauge doppler/frequency shifting.	Demonstration	L1-MIS-015
61	SYS	L2-SYS-061	Downlink Capacity	The CubeSat shall be capable of downlinking at least 7 multispectral images per day.	Data should be returned in a timely manner due to health risks involved with HABs.	Demonstration	L1-SCI-014
71	MIS	L2-MIS-071	Deployment Wait Time	The CubeSat shall not deploy any deployables until at least 30 minutes have elapsed from exiting its dispenser.	Standard waiting time from the CubeSat Design Specification.	Demonstration	L1-PRJ-023
72	MIS	L2-MIS-072	Transmission Wait Time	The CubeSat shall not transmit any signals until at least 45 minutes have elapsed from exiting its dispenser.	Standard waiting time for transmission from CubeSat Design Specification.	Demonstration	L1-PRJ-023
73	MIS	L2-MIS-073	Independent Startup	The CubeSat shall independently perform all startup activities (initialization, detumbling, deployables) within 1 week of exiting its dispenser.	Capable of handling startup activities on its own while out of contact with ground.	Analysis, Test	L1-PRJ-022

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
74	MIS	L2-MIS-074	Imaging Criteria	The CubeSat shall attempt to image a target when the entire feature can be imaged at a GSD below 90 meters per pixel.	Guarantees all parts of an image meet specified resolution requirements.	Analysis, Test	L1-SCI-009
75	MIS	L2-MIS-075	Image Downlink	The CubeSat shall downlink multispectral images from storage upon command by the Ground Station.	Ground driven downlinking allows ground operators to select data to retrieve.	Test	L1-MIS-015
78	MIS	L2-MIS-078	Transmission Shutdown	The CubeSat shall be able to disable and reenable transmissions from any onboard transmission-capable devices via ground command.	Required to prevent interference with other signals on same/similar frequencies.	Demonstration	L1-PRJ-019
79	MIS	L2-MIS-079	Command Encryption	All non-amateur radio commands to the CubeSat shall be encrypted.	Encryption prevents malicious use of satellite communication links.	Inspection	L1-PRJ-018, L1-PRJ-020, L1-PRJ-021
80	MIS	L2-MIS-080	Amateur Repeater	The CubeSat should act as an amateur radio repeater when in nominal mode and not actively imaging or downlinking.	Serves as a contribution to amateur radio arts to justify band usage.	Demonstration	L1-PRJ-018
81	MIS	L2-MIS-081	Semi-Autonomous Operation	The CubeSat shall be able autonomously execute the nominal mission sequence without real-time ground interaction.	Must function on last issued instructions when out of communication range.	Test	L1-MIS-015
82	MIS	L2-MIS-082	Command Acknowledgement & Rejection Reporting	The CubeSat shall transmit an execution confirmation or a command-rejection status with an error reason to the Ground Station for every telecommand received.	Requires positive confirmation of executed or rejected commands.	Demonstration, Test	L1-MIS-015
83	MIS	L2-MIS-083	Time-Tagged Scheduled Command Execution	The CubeSat shall support receiving, queuing, and autonomously executing time-tagged telecommands at ground-specified future epoch timestamps.	Allows uplink of time-tagged commands in advance for execution over target regions.	Demonstration, Test	L1-MIS-015
84	MIS	L2-MIS-084	Verified Handshake Image Purging	The CubeSat shall retain captured image files in non-volatile memory until an explicit purge command is received from the Ground Station.	Prevents loss of science data if downlink pass experiences signal dropouts.	Demonstration, Test	L1-MIS-015

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
129	OBC	L3-OBC-129	Command Decryption	The OBC Subsystem shall decrypt encrypted telecommand	Command packets will be encrypted and will	Demonstration, Test	L2-MIS-079

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
				packets received from the Comms Subsystem prior to validation.	require a decryption process prior to validation and potential execution		
130	OBC	L3-OBC-130	Command Authentication	The OBC Subsystem shall verify the cryptographic authenticity of all non-amateur telecommand packets received by the Comms Subsystem.	Cryptographic authentication prevents unauthorized third parties from injecting malicious commands into the spacecraft.	Demonstration, Test	L2-MIS-079, L2-SYS-049
131	OBC	L3-OBC-131	Command Validation	The OBC Subsystem shall validate the packet structure, parameter bounds, and operational mode compatibility of all received commands before execution.	The satellite must reject malformed, out-of-bounds, or mode-incompatible commands to prevent unintended actions or flight software faults.	Demonstration, Test	L2-SYS-049
132	OBC	L3-OBC-132	Command Status Reporting	The OBC Subsystem should transmit an execution confirmation or a rejection status code to the Ground Station for every telecommand received.	Ground Station would benefit greatly from providing positive confirmation of whether an uplinked telecommand executed successfully or was rejected.	Demonstration, Test	L2-MIS-082
133	OBC	L3-OBC-133	Primary Image Downlink Operation	The OBC Subsystem shall retrieve, package, and stream specified image data to the high-data-rate S-Band radio upon Ground Station command.	The OBC Subsystem coordinates payload data retrieval and framing, utilizing the high-rate S-Band link for full imagery and the UHF link as a low-bandwidth backup for thumbnails and metadata.	Demonstration, Test	L2-SYS-061, L2-MIS-075
134	OBC	L3-OBC-134	Backup Image Downlink Operation	The OBC Subsystem should route compressed image previews or metadata packets to the UHF radio upon Ground Station command.	Provides a redundant, low-bandwidth pathway to retrieve critical image thumbnails and metadata if the primary S-Band link is unavailable.	Demonstration, Test	L2-MIS-075
135	OBC	L3-OBC-135	Telemetry Downlink Operation	The OBC Subsystem shall transmit the most recent full system telemetry data to the Comms Subsystem for downlinking upon Ground Station command.	The OBC subsystem is responsible for all data handling and transmission coordination; the low-data rate link should be sufficient for downlinking telemetry (which will be much smaller than payload data), but failure of the	Demonstration, Test	L2-SYS-043

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
					low data rate link may require use of the high data rate link; the heartbeat beacon will be a reduced set of data, while this command covers the full spectrum of telemetry data.		
136	OBC	L3-OBC-136	Error Log Downlink Operation	The OBC Subsystem should transmit recorded historical fault and error logs to the Comms Subsystem for downlinking upon Ground Station command.	Retrieving historical fault records allows ground operators to diagnose transient anomalies that occurred while the satellite was out of ground contact.	Demonstration, Test	L2-SYS-043
137	OBC	L3-OBC-137	Subsystem Mode Configuration	The OBC Subsystem shall configure the operational state of all onboard subsystems according to the active spacecraft operational mode.	The OBC subsystem is responsible for commanding all other computational devices onboard either directly or indirectly through slave computers, during all phases of the mission; most of this will occur when the satellite is out of ground communication range.	Demonstration, Test	L2-MIS-081
138	OBC	L3-OBC-138	Subsystem Monitoring	The OBC Subsystem shall collect and analyze subsystem telemetry data from all onboard subsystems every 60 seconds.	The OBC subsystem must be aware of the satellite's current state to determine appropriate commands to the other subsystems, this same information would form the basis of telemetry to the ground and heartbeat beacons.	Demonstration, Test	L2-SYS-048, L2-SYS-060
139	OBC	L3-OBC-139	Asynchronous Subsystem Fault Handling	The OBC Subsystem shall process asynchronous fault alerts from onboard subsystems immediately upon receipt.	Critical hardware anomalies such as low battery voltage or payload faults require immediate response rather than waiting for the next 60-second polling cycle.	Demonstration, Test	L2-SYS-048
140	OBC	L3-OBC-140	Heartbeat Operation	The OBC Subsystem shall construct a heartbeat packet and send it to the Comms Subsystem for transmission immediately after collection of subsystem telemetry, if transmissions are allowed and power is sufficient.	Constructing and transmitting the heartbeat immediately after updating telemetry guarantees that the information contained within the signal is up to date.	Demonstration, Test	L2-SYS-060

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
141	OBC	L3-OBC-141	System Timekeeping	The OBC Subsystem shall maintain a system clock and spacecraft epoch time.	The satellite must enforce a 30-minute and 45-minute delays after deployment for transmission and deployment of elements	Demonstration, Test	L2-MIS-071, L2-MIS-072
142	OBC	L3-OBC-142	Primary GPS Time Synchronization	The OBC Subsystem shall synchronize its spacecraft epoch time using GPS time readings provided by the ADCS Subsystem.	Autonomous synchronization against ADCS GPS time readings prevents onboard clock drift over the mission lifetime and ensures accurate image georeferencing and pass execution.	Demonstration, Test	L2-SYS-041, L2-SYS-049
143	OBC	L3-OBC-143	Fallback Ground Time Synchronization	The OBC Subsystem shall synchronize its spacecraft epoch time using Ground Station time-synchronization telecommands when GPS time readings are unavailable.	Provides a reliable ground-commanded fallback to correct onboard clock drift if the ADCS GPS receiver experiences an outage or fails to acquire a lock.	Demonstration, Test	L2-SYS-049
144	OBC	L3-OBC-144	Data Storage	The OBC Subsystem shall have at least TBD GB of onboard data storage.	The OBC needs to be able to store at least enough data to store all telemetry and chlorophyll index data between ground station passes with a \significant buffer.	Inspection, Test	L2-SYS-042
145	OBC	L3-OBC-145	State Non-Volatile Storage	The OBC Subsystem shall provide radiation-tolerant non-volatile memory for storing critical system states, mission phase flags, and inhibit timers.	Critical boot flags and deployment timers must be stored in radiation-tolerant memory so they are not lost or corrupted during power brownouts or resets.	Inspection, Test	L2-SYS-059, L2-MIS-071, L2-MIS-072
146	OBC	L3-OBC-146	Storage Management	The OBC Subsystem shall have the ability to store, compress, decompress, and delete data in storage.	These data management functions are required to maintain enough storage for new data and prepare image data for downlinking.	Demonstration, Test	L2-SYS-042
147	OBC	L3-OBC-147	Verified Image Storage Purging	The OBC Subsystem shall retain captured image files in non-volatile storage on the Payload Subsystem until an explicit Ground Station command is received.	Retaining image files onboard until the Ground Station confirms complete reception prevents permanent loss of science data if a downlink pass drops out.	Demonstration. Test	L2-SYS-042, L2-MIS-084

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
148	OBC	L3-OBC-148	Data Storage Telemetry	The OBC Subsystem shall collect and analyze status and health information from onboard data storage device(s) as part of collecting subsystem telemetry.	Monitoring storage capacity, wear, and block errors is required to prevent silent data corruption or storage exhaustion.	Demonstration, Test	L2-SYS-043
149	OBC	L3-OBC-149	Main Computer Telemetry	The Main Computer of the OBC Subsystem shall collect and analyze its own status and health information as part of collecting subsystem telemetry.	Monitoring the internal resource utilization and health of the central flight computer is necessary to detect software anomalies or thermal issues early.	Demonstration, Test	L2-SYS-043
150	OBC	L3-OBC-150	Data Processor Telemetry	The OBC Subsystem shall collect and analyze health and processing status telemetry from the Data Processor as part of collecting subsystem telemetry.	Tracking the operational status of the secondary image processor ensures processing faults or hangs are detected and logged.	Demonstration, Test	L2-SYS-043
151	OBC	L3-OBC-151	TLE Forwarding	The OBC Subsystem should forward TLEs, target coordinates, and flags to the ADCS Subsystem upon receipt from the Ground Station.	Supplying the ADCS with updated TLEs, target coordinates, and flags enables onboard calculation of upcoming target imaging windows.	Demonstration, Test	L2-SYS-041, L2-MIS-081
152	OBC	L3-OBC-152	Image Constraint Verification	The OBC Subsystem shall verify that all imaging constraints are met prior to imaging a target.	As imaging should occur only under the correct constraints, the OBC Subsystem must be able to actually verify that these criteria are met before issuing an imaging command.	Test	L2-MIS-074
153	OBC	L3-OBC-153	Attitude Mode Commanding	The OBC Subsystem shall command the ADCS Subsystem to transition between operational attitude modes based on the current mission schedule.	Although the ADCS is responsible for actual low-level sensing, state estimation, and control, the OBC Subsystem actually must decide where the satellite is pointing to coordinate the operations of the overall CubeSat.	Demonstration, Test	L2-SYS-038
154	OBC	L3-OBC-154	Imaging Execution	The OBC Subsystem shall command Payload Subsystem imager(s) to capture multispectral images.	As the OBC Subsystem is responsible for verifying imaging constraints are met, it should also be responsible for issuing the command to actually image.	Demonstration	L2-MIS-074

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
155	OBC	L3-OBC-155	Amateur Repeater Operation	The OBC Subsystem shall send a received amateur radio transmission back to the same Comms Subsystem link it was received upon to be rebroadcasted	The OBC Subsystem is responsible for command interpretation and commanding the Comms Subsystem transceivers.	Demonstration	L2-MIS-080
156	OBC	L3-OBC-156	OBC Reset Handling	The OBC Subsystem shall reset itself and all other systems to a pre-launch state when the onboard deployment switch(es) are cycled.	This implements the ground reset functionality but requires interpretation of the action as the from the deployment switches rather than from power loss or brownout.	Demonstration	L2-MIS-074
157	OBC	L3-OBC-157	OBC Standby Power Consumption	The OBC Subsystem shall not exceed a power draw of TBD Watts during Standby Mode.	Constraining continuous standby power consumption of the computing elements ensures the spacecraft maintains a positive orbit-average energy balance.	Test	L2-SYS-046
158	OBC	L3-OBC-158	OBC Peak Power Consumption	The OBC Subsystem shall not exceed a peak power draw of TBD W during active image processing.	Bounding peak power draw when both the Main Computer and Data Processor are active ensures the power distribution bus is not overloaded.	Test	L2-SYS-046
159	OBC	L3-OBC-159	Software Fault Recovery	The OBC Subsystem shall detect and recover from onboard software exceptions, task hangs, and memory corruption events.	Autonomous software fault recovery prevents transient radiation upsets or thread deadlocks from causing permanent loss of the spacecraft.	Test	L2-SYS-049
160	OBC	L3-OBC-160	Communication Bus Fault Recovery	The OBC Subsystem shall detect and isolate communication faults on internal data buses without compromising unaffected subsystems.	Isolating and resetting faulted communication lines prevents a single unresponsive sensor or controller from locking up spacecraft command and data handling.	Test	L2-SYS-048
161	OBC	L3-OBC-161	Safe Mode Transition	The OBC Subsystem shall transition the spacecraft to Safe Mode autonomously upon detecting unrecoverable faults or critical low-battery alerts, or upon Ground Station command.	Safeing the spacecraft in due to unrecoverable faults will help facilitate ground-commanded recovery operations.	Test	L2-SYS-048
162	OBC	L3-OBC-162	Safe Mode Exit	The OBC Subsystem shall transition the spacecraft out of Safe Mode back to nominal operational modes only upon	The spacecraft will only be able to exit safe mode after faults have	Test	L2-SYS-048

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
				receiving an authenticated Ground Station command.	been manually cleared or resolved.		
163	OBC	L3-OBC-163	OBC Radiation Shielding	The OBC Subsystem should include radiation shielding for the Main Computer, Data Processor, and Data Storage components.	The OBC Subsystem is required for successful mission operations and must be resistant to radiation effects, including prevention of data corruption; due to the size of the data several alternative schemes (e.g. redundant copies of data) are impractical.	Analysis, Inspection	L2-SYS-059
164	OBC	L3-OBC-164	OBC Mass Limit	The OBC Subsystem shall not exceed TBD grams in mass.	This mass value has been assigned to the OBC Subsystem based on a decomposition of the satellite's overall mass limit.	Inspection	L2-SYS-027
165	OBC	L3-OBC-165	OBC Volume	All elements of the OBC Subsystem shall fit within a volume of 95 mm x 95 mm x 25 mm, with the exception of external interfaces.	From a preliminary volume budget based on initial preliminary design and volume of the prospective required components	Inspection	L2-SYS-026
166	OBC	L3-OBC-166	Radiation-Resistant Reboot	The OBC Subsystem should be capable of overwriting its flight software from a radiation resistant memory device during a reboot	This would allow the OBC subsystem to pull a fresh uncorrupted copy of the software in the case of radiation-related corruption of the flight software.	Demonstration	L2-SYS-059
167	OBC	L3-OBC-167	On-Board Chlorophyll Index Processing	The OBC Subsystem should include a dedicated Data Processor capable of executing on-board Chlorophyll Index conversion algorithms on captured multispectral imagery.	Offloading compute-intensive multispectral image processing to a dedicated secondary processor protects real-time scheduling on the Main Computer. This offers the potential path of spending down only these images once battle tested	Demonstration, Test	L2-SYS-061, L2-MIS-081
168	OBC	L3-OBC-168	Schedule Telecommand Execution	The OBC Subsystem shall store time-tagged telecommands in a schedule queue and autonomously execute them when their specified spacecraft epoch timestamp is reached.	Enables the Ground Station to schedule mission operations that must occur when the satellite is outside of Ground Station visibility.	Demonstration, Test	L2-SYS-049, L2-MIS-081, L2-MIS-083

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
169	OBC	L3-OBC-169	Post-Deployment Inhibit Enforcement	The OBC Subsystem shall enforce a minimum 30-minute wait timer before actuating deployables and a minimum 45-minute wait timer before enabling RF transmissions after dispenser ejection.	Ensures strict compliance with CubeSat launch dispenser safety hold times prior to physical deployment or RF emission.	Demonstration, Test	L2-MIS-071, L2-MIS-072
170	OBC	L3-OBC-170	Autonomous Startup Sequencing	The OBC Subsystem shall autonomously coordinate deployable release and initial detumbling initialization within 1 week of exiting the dispenser.	Guarantees the satellite can independently initialize, deploy mechanisms, and establish a stable power-positive attitude prior to initial ground contact.	Demonstration, Test	L2-SYS-038, L2-MIS-073
171	OBC	L3-OBC-171	Commanded RF Transmission Inhibit	The OBC Subsystem shall disable and reenable all onboard RF transmissions upon receiving a valid Ground Station transmission shutdown command.	Required by spectrum regulatory authorities to immediately cease spacecraft RF emissions if interference occurs.	Demonstration, Test	L2-SYS-049, L2-MIS-078
172	OBC	L3-OBC-172	Integrated Ground Diagnostic Interface	The OBC Subsystem shall provide an external umbilical interface accessible when the CubeSat is fully integrated to support software flashing and diagnostic testing.	Allows engineers to update flight software, reset pre-launch states, and verify subsystem health after final mechanical integration.	Inspection, Test	L2-SYS-053
173	OBC	L3-OBC-173	Image Georeference Metadata Tagging	The OBC Subsystem shall associate synchronized spacecraft ECEF position coordinates and LVLH attitude estimates with each captured image set.	Binding orbital position and orientation state vectors to each image at capture time is required to geolocate harmful algal blooms and satisfy mission science objectives.	Demonstration, Test	L2-SYS-040, L2-SYS-041
174	OBC	L3-OBC-174	Thermal Control Actuation	The OBC Subsystem shall enforce subsystem thermal operating inhibits based on collected temperature telemetry.	Thermal operational cutoffs protect temperature-sensitive components such as batteries and optical sensors from permanent damage.	Demonstration, Test	L2-SYS-050
175	OBC	L3-OBC-175	On-Orbit Parameter Reconfiguration	The OBC Subsystem should support updating configurable operational thresholds, timeouts, and algorithm coefficients on orbit via Ground Station telecommands.	Enables Ground Station to tune control loop tolerances, safety cutoffs, and science processing constants after launch without reflashing flight software.	Demonstration, Test	L2-SYS-049
176	OBC	L3-OBC-176	On-Orbit Flight Software	The OBC Subsystem should support receiving, verifying, and installing updated flight software	Provides the ability to patch software defects or upgrade onboard	Demonstration, Test	L2-SYS-044, L2-SYS-049

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
			Reprogrammability	images on orbit via Ground Station telecommands.	processing logic during the orbital mission while maintaining fallback recovery to the golden image.		
185	OBC	L4-OBC-185	Telecommand Decryption	The OBC shall decrypt encrypted telecommand frames prior to command parsing.	Protects command contents from unauthorized disclosure over the space-to-ground RF link.	Test	L3-OBC-129
186	OBC	L4-OBC-186	Telecommand Authentication	The OBC shall verify the cryptographic authentication tag of each non-amateur telecommand frame prior to command execution.	Prevents unauthorized transmitters from injecting malicious commands into the spacecraft.	Test	L3-OBC-130
187	OBC	L4-OBC-187	Telecommand Replay Protection	The OBC shall reject non-amateur telecommand frames that contain a duplicate or out-of-window sequence counter.	Prevents an adversary from capturing and re-transmitting previously valid telecommands.	Test	L3-OBC-130
188	OBC	L4-OBC-188	Cryptographic Key Isolation	The OBC shall store cryptographic keys in write-protected non-volatile memory inaccessible to unauthenticated interfaces.	Prevents accidental overwrite or unauthorized readback of command encryption keys.	Inspection, Test	L3-OBC-129, L3-OBC-130
189	OBC	L4-OBC-189	Unauthenticated Frame Rejection	The OBC shall discard telecommand frames that fail decryption or authentication checks and increment a security rejection counter.	Ensures corrupted or spoofed frames are never passed to the command dispatcher while providing visibility into link tampering.	Test	L3-OBC-129, L3-OBC-130
190	OBC	L4-OBC-190	Command Syntax Validation	The OBC shall validate the packet length, structure, and error-detection checksum of each received telecommand frame against the command dictionary.	Rejects frames corrupted by RF link noise and prevents malformed packets from causing buffer overflows.	Test	L3-OBC-129
191	OBC	L4-OBC-191	Command Parameter Bounds Checking	The OBC shall verify that all command arguments fall within defined operational boundary limits prior to command dispatch.	Prevents out-of-range parameters from causing arithmetic faults or unsafe subsystem actuation.	Test	L3-OBC-129
192	OBC	L4-OBC-192	Command Mode Authorization	The OBC shall reject telecommands that are not authorized for execution in the active spacecraft operational mode.	Prevents ground commands from triggering high-power or unsafe actions while the spacecraft is in restricted modes.	Test	L3-OBC-129
193	OBC	L4-OBC-193	Critical Command Safeguard	The OBC shall require a secondary confirmation check prior to executing commands	Reduces the risk of accidental data loss, unintended radio	Test	L3-OBC-129, L3-OBC-156

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
				that reboot processors, cycle power rails, inhibit RF transmissions, enter Emergency Mode, or delete stored mission data.	silence, or disruptive reboots from a single erroneous command.		
194	OBC	L4-OBC-194	Command Execution Acknowledgment	The OBC shall generate and queue a positive acknowledgment packet for every telecommand that passes validation and executes.	Provides ground operators with immediate confirmation that a transmitted command succeeded.	Demonstration, Test	L3-OBC-129
195	OBC	L4-OBC-195	Command Rejection Reporting	The OBC shall generate and queue a negative acknowledgment packet containing an error reason code for every rejected telecommand.	Enables ground operators to diagnose why a command was rejected and resend a corrected command during the pass.	Demonstration, Test	L3-OBC-129
196	OBC	L4-OBC-196	Command Outcome Logging	The OBC shall record the execution or rejection status of every received telecommand in non-volatile log storage.	Preserves an onboard audit trail of all received commands for post-pass verification and anomaly investigation.	Test	L3-OBC-129, L3-OBC-131
197	OBC	L4-OBC-197	Scheduled Command Queueing	The OBC shall store valid telecommands tagged with a future execution time in a fixed-capacity schedule queue.	Allows operators to uplink time-delayed commands during a ground pass for later execution over target regions.	Demonstration, Test	L3-OBC-154
198	OBC	L4-OBC-198	Scheduled Command Dispatch	The OBC shall autonomously dispatch queued time-tagged telecommands when the spacecraft epoch clock reaches their specified execution time.	Ensures scheduled operations execute deterministically without requiring real-time ground contact.	Demonstration, Test	L3-OBC-135, L3-OBC-154
199	OBC	L4-OBC-199	Downlink Frame Segmentation	The OBC shall segment image and telemetry files into sequential transfer frames containing sequence numbers and frame error-detection codes.	Enables the Ground Station to detect missing frames and reassemble large files across noisy RF links.	Test	L3-OBC-130, L3-OBC-131
200	OBC	L4-OBC-200	High-Rate Downlink Flow Control	The OBC shall regulate data transfer to the high-data-rate downlink transmitter based on transmitter buffer status feedback.	Prevents transmitter buffer overflows and underruns during high-speed S-Band image downlinks.	Test	L3-OBC-130
201	OBC	L4-OBC-201	Low-Rate Image Preview Routing	The OBC shall support routing compressed image previews and image metadata packets to the low-data-rate telemetry transceiver upon command.	Provides a backup path to verify image capture and retrieve thumbnails if the primary high-rate transmitter fails.	Demonstration, Test	L3-OBC-130
202	OBC	L4-OBC-202	On-Demand Telemetry Packaging	The OBC shall compile current subsystem health data, historical telemetry logs, and	Allows ground operators to retrieve comprehensive health	Demonstration, Test	L3-OBC-131

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
				recorded fault flags into transfer frames upon Ground Station request.	histories and fault logs during a communication pass.		
203	OBC	L4-OBC-203	Housekeeping Beacon Assembly	The OBC shall assemble an unencrypted housekeeping beacon packet containing spacecraft time, operational mode, battery voltage, battery State of Charge, temperatures, reboot count, and fault flags.	Provides a compact summary of essential survival metrics for rapid ground assessment and amateur radio reception.	Test, Demonstration	L3-OBC-134
204	OBC	L4-OBC-204	Housekeeping Beacon Cadence	The OBC shall dispatch the housekeeping beacon packet at a fixed 49-second interval whenever RF transmissions are enabled and no active downlink is in progress.	Establishes a predictable beacon period for ground tracking while avoiding collisions with active data downlinks.	Test, Demonstration	L3-OBC-134
205	OBC	L4-OBC-205	Beacon Low-Power Suppression	The OBC shall automatically suspend housekeeping beacon transmissions when the spacecraft is in Low-Power Mode or when RF Inhibit is active.	Conserves battery energy during power-negative recovery states and enforces regulatory radio silence when commanded.	Test	L3-OBC-134, L3-OBC-156
206	OBC	L4-OBC-206	Uplink Reception Priority	The OBC shall prioritize incoming telecommand reception over outgoing beacon or repeater transmissions on shared half-duplex transceiver channels.	Guarantees that ground commandability is never blocked by outgoing housekeeping or amateur radio traffic.	Test	L3-OBC-129, L3-OBC-143
207	OBC	L4-OBC-207	Amateur Repeater Buffering	The OBC shall stage valid incoming amateur radio repeater packets in a dedicated circular ring buffer of at least 64 kilobits capacity.	Isolates public amateur repeater traffic in a bounded memory space so it cannot encroach on mission data buffers.	Test	L3-OBC-143
208	OBC	L4-OBC-208	Amateur Repeater Rate Limiting	The OBC shall enforce a maximum transmission duty cycle on rebroadcast amateur radio repeater packets.	Prevents high-volume amateur radio use from overheating the UHF transceiver or depleting the battery.	Test	L3-OBC-143
209	OBC	L4-OBC-209	Operational Mode State Machine	The OBC shall execute a deterministic Finite State Machine governing transitions across Pre-Launch, Initialization, Checkout, Standby, Imaging, Downlink, Repeater, Low-Power, Safe, and Emergency modes.	Ensures the spacecraft operates in a single well-defined state at all times with deterministic mode transition rules.	Test, Demonstration	L3-OBC-132, L3-OBC-147, L3-OBC-148
210	OBC	L4-OBC-210	Subsystem Power Rail Commanding	The OBC shall command the Power Subsystem to enable or disable individual subsystem power rails according to the active operational mode.	Ensures only the subsystems required for the current operational mode draw electrical power.	Test, Demonstration	L3-OBC-132

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
211	OBC	L4-OBC-211	Graceful Subsystem Shutdown	The OBC shall issue a shutdown notification to active subsystems and verify completion of pending data writes prior to commanding removal of their power rails.	Prevents flash file system corruption and hardware transients during nominal mode transitions.	Test	L3-OBC-132, L3-OBC-137
212	OBC	L4-OBC-212	Non-Volatile State Persistence	The OBC shall record the active operational mode, mission phase flags, boot counter, and cumulative inhibit timers in radiation-tolerant non-volatile memory at least once every 30 seconds.	Preserves spacecraft state across unexpected power brownouts or processor resets so initialization sequences are not repeated.	Test	L3-OBC-132, L3-OBC-136, L3-OBC-155
213	OBC	L4-OBC-213	Post-Deployment Inhibit Timers	Upon initial boot following dispenser ejection, the OBC shall enforce a 30-minute cumulative uptime delay before actuating deployables and a 45-minute delay before enabling RF transmissions.	Complies with CubeSat launch safety hold requirements prior to physical deployment or RF emission.	Test, Demonstration	L3-OBC-135, L3-OBC-155
214	OBC	L4-OBC-214	Deployable Actuation Current Limiting	The OBC shall limit release mechanism actuation current to a maximum of 2.5 A during deployable release sequences.	Prevents release mechanism activation from causing overcurrent trips on the spacecraft power bus.	Test	L3-OBC-155
215	OBC	L4-OBC-215	Deployable Feedback Debouncing	The OBC shall filter deployable status switch signals over a 100 ms debounce window and retry release actuation up to 2 times if deployment is unconfirmed.	Eliminates false deployment indications caused by mechanical switch bounce and provides retry capability if a mechanism sticks.	Test	L3-OBC-155, L3-STR-199
216	OBC	L4-OBC-216	Pre-Launch Reset Switch Handling	When the physical deployment switches are cycled during ground testing, the OBC shall zero all persisted post-deployment timers and clear mission phase flags in non-volatile memory.	Guarantees the spacecraft launches with clean inhibit timers and pre-launch flags after integrated ground testing.	Test, Demonstration	L4-OBC-144
217	OBC	L4-OBC-217	Monotonic Clock Maintenance	The OBC shall maintain a continuous monotonic tick counter and a Real-Time Clock with a drift rate not exceeding 20 ppm.	Provides a continuous time base for relative timeout evaluation and orbital event scheduling.	Test	L3-OBC-135
218	OBC	L4-OBC-218	Primary GPS Time Synchronization	The OBC shall discipline its spacecraft epoch clock using GPS time messages and synchronization pulses received from the ADCS Subsystem.	Maintains long-term onboard clock accuracy autonomously without requiring ground time updates.	Test, Demonstration	L3-OBC-135
219	OBC	L4-OBC-219	Fallback Ground Time Synchronization	When GPS time synchronization is unavailable, the OBC shall update its spacecraft epoch clock upon receiving a valid	Ensures onboard clock accuracy can still be maintained via ground command if the GPS	Test, Demonstration	L3-OBC-135

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
				time-synchronization telecommand from the Ground Station.	receiver experiences an outage.		
220	OBC	L4-OBC-220	Subsystem Telemetry Polling Scheduler	The OBC shall poll health and status telemetry from the Power, ADCS, Comms, Thermal, and Payload subsystems at a fixed 60-second period.	Provides deterministic, periodic visibility into all subsystem health states.	Test, Demonstration	L3-OBC-133
221	OBC	L4-OBC-221	Telemetry Threshold Evaluation	The OBC shall compare collected subsystem telemetry values against mode-specific high and low operational bounds and assert a fault flag when a bound is exceeded.	Enables automated detection of off-nominal voltages, currents, or temperatures before hardware damage occurs.	Test	L3-OBC-133, L3-OBC-147
222	OBC	L4-OBC-222	Asynchronous Fault Alert Handling	The OBC shall preempt background tasks to process critical asynchronous fault alerts from onboard subsystems within 100 ms of receipt.	Ensures rapid protective action when a subsystem reports an acute hardware fault or low-battery emergency.	Test	L3-OBC-133, L3-OBC-147
223	OBC	L4-OBC-223	Lossless Image Compression	The OBC shall compress captured multispectral and truth image files using a lossless compression algorithm prior to downlink packaging.	Reduces image file sizes for downlink without altering radiometric pixel values needed for scientific algorithms.	Test	L3-OBC-137
224	OBC	L4-OBC-224	Stored File Integrity Verification	The OBC shall compute and store a cyclic redundancy check (CRC) value for every file written to non-volatile mass storage and verify it upon readback.	Detects bit-flip corruption in stored science imagery and telemetry logs prior to downlink.	Test	L3-OBC-136, L3-OBC-137
225	OBC	L4-OBC-225	Ground-Verified Image Deletion	The OBC shall delete stored image files and clear associated camera buffers only upon receiving an authenticated image purge command from the Ground Station.	Prevents permanent science data loss if a downlink pass is interrupted before the Ground Station receives the complete image.	Test, Demonstration	L3-OBC-137
226	OBC	L4-OBC-226	Storage Capacity Telemetry	The OBC shall measure and report available free capacity on non-volatile mass storage during each 60-second telemetry collection cycle.	Allows ground operators and flight software to track storage consumption across imaging passes.	Test	L3-OBC-138
227	OBC	L4-OBC-227	Flash Bad-Block Remapping	The OBC shall automatically remap write operations away from defective non-volatile memory blocks and increment a bad-block telemetry counter.	Prevents localized flash memory wear or radiation damage from halting data storage operations.	Test	L3-OBC-137, L3-OBC-138
228	OBC	L4-OBC-228	Storage Wear-Leveling Telemetry	The OBC shall distribute write operations across non-volatile mass memory blocks and report	Extends the operational lifespan of flash storage over two years of	Inspection, Test	L3-OBC-138

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
				memory wear-leveling status in storage telemetry.	continuous orbital logging and imaging.		
229	OBC	L4-OBC-229	High Storage Utilization Warning	The OBC shall assert a storage warning telemetry flag when non-volatile mass storage exceeds 85% of total capacity.	Alerts ground operators to prioritize downlinking and purging stored images before memory fills.	Test	L3-OBC-138
230	OBC	L4-OBC-230	Telemetry Log FIFO Rollover	When non-volatile mass storage reaches 99% capacity, the OBC shall overwrite the oldest unflagged historical telemetry logs in First-In, First-Out order while preserving unpurged science images.	Guarantees continuous health logging during extended communication outages without overwriting science imagery.	Test	L3-OBC-137, L3-OBC-138
231	OBC	L4-OBC-231	Processor Temperature Telemetry	The OBC shall sample and report the internal temperatures of the Main Computer and Data Processor during each 60-second telemetry collection cycle.	Enables thermal monitoring and fault detection for the onboard computing elements.	Test	L3-OBC-139
232	OBC	L4-OBC-232	Processor Voltage and Current Telemetry	The OBC shall sample and report the supply voltages and current consumption of the Main Computer and Data Processor during each 60-second telemetry collection cycle.	Tracks computing power consumption and detects voltage rail brownouts or anomalous current draw.	Test	L3-OBC-139
233	OBC	L4-OBC-233	Processor Resource Utilization Telemetry	The OBC shall record and report CPU load, RAM utilization, uptime, and reboot counters for the Main Computer and Data Processor during each 60-second telemetry collection cycle.	Provides visibility into software resource margins and unexpected processor resets.	Test	L3-OBC-139
234	OBC	L4-OBC-234	Ephemeris Target Forwarding	Upon receiving updated TLEs or target coordinates from the Ground Station, the OBC shall forward them to the ADCS Subsystem and request an updated pass schedule.	Ensures the ADCS orbit propagator calculates imaging windows using the latest orbital elements and target locations.	Test, Demonstration	L3-OBC-140, L3-OBC-141
235	OBC	L4-OBC-235	Upcoming Pass Schedule Queue	The OBC shall maintain a fixed-capacity queue of up to 16 upcoming target imaging passes received from the ADCS Subsystem.	Ensures the ADCS orbit propagator calculates imaging windows using the latest orbital elements and target locations.	Test	L3-OBC-140
236	OBC	L4-OBC-236	Pass Schedule Conflict Resolution	The OBC shall discard duplicate pass entries and resolve overlapping imaging pass windows by selecting the pass with higher priority or maximum elevation.	Prevents conflicting attitude slews when multiple target windows overlap.	Test, Demonstration	L3-OBC-140

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
237	OBC	L4-OBC-237	Pre-Pass Constraint Verification	Prior to initiating an imaging pass sequence, the OBC shall verify that battery State of Charge is at least 20%, target ground sampling distance is below 90 meters per pixel, and no critical faults are active.	Prevents high-power camera activation when power reserves or orbital pass geometry are insufficient.	Test, Analysis	L3-OBC-140
238	OBC	L4-OBC-238	Pre-Pass Target Preparation	At the scheduled lead time prior to target acquisition, the OBC shall enable the Payload power rail, command the ADCS to track the target, and command the Payload to begin sensor warmup.	Ensures the camera sensors are thermally stabilized and pointed at the target upon entering the imaging window.	Test, Demonstration	L3-OBC-141, L3-OBC-142
239	OBC	L4-OBC-239	Synchronized Capture Triggering	The OBC shall command the Payload to trigger image capture only after receiving both target-lock confirmation from the ADCS and warmup-ready confirmation from the Payload within the valid pass window.	Prevents blurred or uncalibrated images by gating shutter activation on both attitude lock and sensor readiness.	Test, Demonstration	L3-OBC-142
240	OBC	L4-OBC-240	Post-Pass Standby Return	Upon receiving capture completion confirmation from the Payload, the OBC shall power down the Payload imaging rail and command the ADCS to return to its standby attitude profile.	Minimizes payload power consumption and orbital drag immediately after image collection finishes.	Test, Demonstration	L3-OBC-141, L3-OBC-142
241	OBC	L4-OBC-241	Imaging Pass Fault Abort	The OBC shall immediately abort an active imaging pass, power down the Payload rail, and halt target tracking if sensor warmup times out, a payload fault occurs, attitude lock times out, or a low-battery alert triggers.	Protects battery reserves and payload hardware if any required imaging condition fails mid-pass.	Test	L3-OBC-140, L3-OBC-142, L3-OBC-146
242	OBC	L4-OBC-242	Ground Pass Preemption	The OBC shall immediately preempt an active imaging pass if an incoming Ground Station telecommand session is detected.	Ensures real-time Ground Station commandability takes precedence over routine autonomous imaging.	Test	L3-OBC-129, L3-OBC-142
243	OBC	L4-OBC-243	Monotonic Pass Timeout Evaluation	The OBC shall evaluate all in-pass phase timeouts using the monotonic tick counter rather than wall-clock epoch time subtraction.	Prevents sudden epoch clock adjustments during GPS or ground time synchronization from causing false timeout aborts mid-pass.	Test	L3-OBC-135, L3-OBC-142
244	OBC	L4-OBC-244	Chloropyhll Index Computation	Following image capture, the OBC shall command the Data Processor to ingest the raw multispectral frames and compute the Chloropyhll Index map.	Generates processed harmful algal bloom data products onboard while keeping the Main Computer free for flight control.	Test, Demonstration	L3-OBC-153

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
245	OBC	L4-OBC-245	Raw Frame Transfer Retry	The Data Processor shall verify the checksum of each raw image block transferred from the Payload and retry corrupted block transfers up to 3 times.	Prevents transient serial link noise from corrupting raw multispectral image transfers.	Test	L3-OBC-146, L3-OBC-153
246	OBC	L4-OBC-246	Image Processing Arithmetic Guard	The Data Processor shall detect arithmetic divide-by-zero or overflow exceptions during Chloropyhll Index computation and fall back to storing the raw uncompressed image.	Guarantees that anomalous pixel values do not crash the processing pipeline or result in lost science imagery.	Test	L3-OBC-146, L3-OBC-153
247	OBC	L4-OBC-247	Preemptive Priority Task Scheduling	The OBC flight software shall execute tasks using preemptive priority-based scheduling so that critical fault-handling and control tasks immediately preempt background tasks.	Guarantees deterministic response times for attitude control, command reception, and fault recovery.	Inspection, Test	L3-OBC-132, L3-OBC-146
248	OBC	L4-OBC-248	Static Memory Allocation	All OBC flight software tasks, queues, and buffers shall allocate memory statically at initialization with zero dynamic heap allocation during flight operations.	Eliminates heap fragmentation, memory leaks, and non-deterministic allocation delays over the two-year mission.	Inspection, Test	L3-OBC-146
249	OBC	L4-OBC-249	Memory Utilization Protection	The OBC shall suspend lowest-priority background tasks if runtime RAM utilization exceeds 95% of available memory.	Prevents stack overflows and kernel crashes under anomalous processing loads.	Test	L3-OBC-139, L3-OBC-146
250	OBC	L4-OBC-250	Hardware Watchdog Reset	The OBC shall strobe an independent hardware watchdog timer during nominal execution that automatically resets the processor if software execution hangs.	Recovers the flight computer automatically from thread deadlocks or infinite loops.	Test, Demonstration	L3-OBC-146
251	OBC	L4-OBC-251	Golden Image Boot Recovery	The OBC bootloader shall verify the integrity of the primary flight software image upon boot and boot from a write-protected fallback image if verification fails across 3 consecutive boot attempts.	Ensures the spacecraft can recover from radiation-induced corruption of the primary flight software memory.	Test, Demonstration	L3-OBC-146, L3-OBC-152
252	OBC	L4-OBC-252	Internal Bus Fault Recovery	The OBC shall retry failed internal bus transactions up to 3 times and automatically reset the local bus controller upon detecting a bus-off or transaction timeout fault.	Isolates transient electrical bus glitches and prevents a hung peripheral from locking up internal communication threads.	Test	L3-OBC-146, L3-OBC-158
253	OBC	L4-OBC-253	Concurrent Storage Access Serialization	The OBC shall serialize concurrent read and write requests to non-volatile storage using mutual exclusion locking and buffer logs in RAM if the	Prevents data corruption from simultaneous storage writes by multiple software tasks.	Test	L3-OBC-137, L3-OBC-146, L3-OBC-158

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
				storage bus is temporarily busy or faulted.			
254	OBC	L4-OBC-254	Autonomous Safe Mode Entry	The OBC shall autonomously transition the spacecraft to Safe Mode and shed non-essential loads if an unrecoverable subsystem fault, persistent bus failure, or critical low-battery alert occurs.	Places the spacecraft in a power-positive, thermally safe, and commandable posture until ground operators can intervene.	Test, Demonstration	L4-OBC-147
255	OBC	L4-OBC-255	Authenticated Safe Mode Exit	The OBC shall clear latched Safe Mode fault flags and return to nominal operational modes only upon receiving a valid Ground Station exit command.	Ensures the spacecraft does not autonomously resume high-power operations before ground operators have diagnosed the cause of Safe Mode entry.	Test, Demonstration	L3-OBC-148
256	OBC	L4-OBC-256	Off-Target Hardware-Mocked Unit Testing	All OBC flight software components shall pass automated off-target unit and integration tests using simulated hardware interfaces with at least 80% statement coverage prior to flight integration.	Verifies software logic, fault handling, and boundary conditions early without requiring physical avionics hardware.	Test, Inspection	L3-OBC-146, L3-OBC-160
257	OBC	L4-OBC-257	Thermal Setpoint Regulation	The OBC shall evaluate component temperature telemetry against high and low thermal setpoints every 60 seconds and command active heater rails or thermal operational inhibits when temperatures cross defined thresholds.	Protects temperature-sensitive subsystems such as batteries and imagers from operating or charging outside their rated thermal limits.	Test, Demonstration	L3-OBC-138, L3-OBC-174
258	OBC	L4-OBC-258	Power Subsystem Watchdog Heartbeat	The OBC shall transmit a periodic communication heartbeat to the Power Subsystem within the Power Subsystem's watchdog timeout window during nominal operations.	Prevents the Power Subsystem from executing an unintended hard power-cycle of the Main Computer or triggering emergency beacon failover during nominal operation.	Test, Demonstration	L3-OBC-137, L3-OBC-138
259	OBC	L4-OBC-259	Onboard Auxiliary Sensor Readout	The OBC should sample its board-level accelerometer and magnetometer sensors during telemetry collection to provide auxiliary body-rate and magnetic field measurements.	Provides an independent backup sensor source to cross-check ADCS attitude rates and verify detumbling progress.	Test	L3-OBC-149
260	OBC	L4-OBC-260	Image Georeference Sidecar Generation	Upon completion of an image capture sequence, the OBC shall query the synchronized ECEF position and LVLH attitude state vectors from the ADCS and record them in the image metadata header.	Embeds accurate geographic center coordinates and pointing orientation with every image to satisfy science co-registration requirements.	Test, Demonstration	L3-OBC-154, L3-OBC-173

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
261	OBC	L4-OBC-261	Data Processor Power Gating	The Main Computer shall keep the Data Processor power rail disabled during Standby Mode, enabling it only during active image post-processing and high-rate S-Band downlink sessions.	Prevents the high-performance secondary processor from drawing idle power during nominal standby orbits.	Test, Demonstration	L3-OBC-137, L3-OBC-157, L3-OBC-167
262	OBC	L4-OBC-262	Data Processor Execution Timeout	The Main Computer shall enforce a maximum execution timer on Data Processor operations, aborting the task and disabling the Data Processor power rail if processing fails to complete within the allocated window.	Prevents a hung image processing job from indefinitely draining spacecraft battery power.	Test	L3-OBC-158, L3-OBC-159, L3-OBC-167
263	OBC	L4-OBC-263	Faulted Bus Node Isolation	If an internal shared communication bus remains unresponsive after 3 local controller reset attempts, the OBC shall sequentially command the Power Subsystem to disable non-essential subsystem rails to isolate the faulted node.	On a shared multi-drop bus such as CAN, a single latched peripheral node can pull the entire bus dominant; isolating non-essential nodes restores critical communications with the Power and ADCS subsystems.	Test	L3-OBC-160, L3-OBC-161
264	OBC	L4-OBC-264	Non-Volatile Parameter Table Management	The OBC shall store configurable operational thresholds, timeouts, and calibration coefficients in a non-volatile parameter table that can be updated individually via telecommand and restored to default values upon command.	Allows low-bandwidth tuning of mission thresholds and algorithm weights in a single pass without risking a full software reflash.	Test, Demonstration	L3-OBC-175
265	OBC	L4-OBC-265	Chunked Software Image Staging	The OBC should receive segmented flight software, update binaries over the telecommand link, assemble them in non-volatile staging memory, and report missing segment indices to the Ground Station.	Enables reliable upload of flight software patches across multiple short Ground Station passes.	Test, Demonstration	L3-OBC-176
266	OBC	L4-OBC-266	Staged Software Signature Verification	The OBC bootloader should verify the cryptographic signature and cyclic redundancy check of a staged software image prior to installation and automatically roll back to the golden image if the new image fails to receive a ground confirmation command within 24 hours.	Prevents corrupted or faulty software updates from permanently disabling spacecraft communications or operations.	Test, Demonstration	L3-OBC-166, L3-OBC-176
268	OBC	L4-OBC-267	Umbilical Diagnostic	The OBC shall provide an external wired serial diagnostic	Enables ground engineers to run	Inspection, Test	L3-OBC-172

No.	Owner	ID	Name	Description	Rationale	Verification Method	Parents
			Command Interface	interface that supports real-time telemetry readout, manual command injection, and pre-launch state verification when the spacecraft is fully integrated.	checkout tests and inspect spacecraft state after final mechanical integration without using RF links.		
268	OBC	L4-OBC-268	Umbilical Flight Software Flashing	The OBC shall expose hardware programming lines through the external umbilical interface to allow reflashing the Main Computer and Data Processor while the spacecraft is fully integrated.	Allows late-stage software fixes prior to dispenser integration without disassembling the CubeSat structure.	Inspection, Test	L3-OBC-172

Classification & Scope Boundaries (Opt-Out of MoSCoW Summary)

We have opted out of the MoSCoW summary in favor of our modal-verb requirement formatting ("shall" vs. "should") across the L0–L4 requirement hierarchy. To maintain strict scope governance for our release and ensure excluded capabilities cannot creep back in, our Assignment 9 requirements map to the following scope boundaries:

- **Mandatory Requirements ("... shall ..."):** All "shall" requirements in the Assignment 9 baseline (L3-OBC-129 through L3-OBC-131, L3-OBC-133, L3-OBC-135, L3-OBC-137–150, L3-OBC-152–162, L3-OBC-164–165, L3-OBC-168–174, L4-OBC-185–258, L4-OBC-260–264, L4-OBC-267–268). Failure to satisfy any "shall" requirement constitutes system failure.
- **Recommended Requirements ("... should ..."):** All "should" requirements in the Assignment 9 baseline (L3-OBC-132, L3-OBC-134, L3-OBC-136, L3-OBC-151, L3-OBC-163, L3-OBC-166, L3-OBC-167, L3-OBC-175, L3-OBC-176, L4-OBC-259, L4-OBC-265, L4-OBC-266). These are strongly recommended and designed into the architecture, though the project will not fail if deferred.
- Explicit Out-of-Scope / Excluded Features ("Won't" List for This Release — Recorded to Prevent Scope Creep):
    1. **No Real-Time SGP4 Orbit Propagation on the Main Computer:** Orbital propagation is computationally expensive and runs exclusively on the ADCS Computer (CMD_ADCS_REQ_SCHEDULE / MSG_ADCS_SCHEDULE_PACKET) and Ground Station; it will not run on the Main Computer.
    2. **No Relational (SQL) Database on the Embedded Flight Computer:** Relational databases introduce dynamic memory overhead and non-deterministic latency that violate hard real-time loop deadlines; onboard storage uses a simpler register/file store in SPI MRAM and NAND Flash (SQLite is restricted strictly to the Ground Station PC).
    3. **No Direct ADCS-to-Imager Triggering Bypassing the Main Computer:** Direct triggering would bypass the centralized SoC (>= 20%), GSD (<= 90 m), and fault checks (L4-OBC-237) managed by the Main Computer.
    4. **No Dynamic Heap Memory Allocation (malloc, new) During Flight Operations:** Strictly prohibited by L4-OBC-248 to prevent heap fragmentation and out-of-memory kernel panics.
    5. **No Proprietary Flight OS or Closed Toolchains:** Commercial platforms such as VxWorks are ruled out by CON-01 and CON-02.

Inherited Constraints (from Week 3 Essay)

Constraint ID	Category	Constraint Statement (from Week 3 Essay)
CON-01	Economic	Due to the financial limitations of the CubeCats organization, we are limited to their already purchased hardware (EnduroSat OBC Type I board) and budget for this project.
CON-02	Economic	Because of financial limitations, we will focus on using open-source tooling (NASA F', FreeRTOS, GCC/CMake, Python) and will not use proprietary software.
CON-03	Professional	This project must undergo a Preliminary Design Review (PDR) by professionals in the aerospace industry as part of the overarching project lifecycle.
CON-04	Professional	Since we are operating within an organization at the University of Cincinnati, our actions and work will reflect the public opinion of the school and the CubeCats organization.
CON-05	Professional	The flight software shall follow established aerospace software engineering practices for requirements management, version control, testing, documentation, and verification.
CON-06	Legal	Comply with all US regulations and procedures relating to export control (ITAR and EAR) and research security (L1-PRJ-021).
CON-07	Legal	Comply with all US regulations and procedures for the utilization of radio frequencies (FCC) and remote sensing (NOAA) (L1-PRJ-019, L1-PRJ-020).
CON-08	Security	Payload data and non-amateur telecommands must be encrypted on downlinks/uplinks to prevent foreign or unauthorized actors from accessing sensitive data or spoofing commands (L2-MIS-079, L3-OBC-129).
CON-09	Security	Critical spacecraft commands shall require additional validation before execution to reduce the risk of accidental or malicious commands (L4-OBC-193).
CON-10	Security	The flight software shall isolate faults within individual subsystems to prevent a failure in one subsystem from unnecessarily compromising the operation of the OBC or other spacecraft subsystems (L3-OBC-160, L4-OBC-263).

Assumptions and Open Questions

ID	Type	Description	Related Requirement IDs	Owner	Target Resolution Date
ASM-01	Assumption	The already-purchased EnduroSat OBC Type I board and STM32F407VG development boards remain electrically functional for hardware-in-the-loop (HITL) testing.	CON-01, L3-OBC-165	Matthew Goldsberry	Verified 2026-10-20 (Ongoing through 2027-01-29)
ASM-02	Assumption	The ADCS Subsystem Computer provides pre-computed target pass windows (MSG_ADCS_SCHEDULE_PACKET, 0x43) every 30 minutes and 1PPS GPS time synchronization pulses (L4-OBC-218).	L3-OBC-142, L3-OBC-151, L4-OBC-218, L4-OBC-235	Elmoataz Abdellah	2026-11-15
ASM-03	Assumption	Maximum encrypted radio packet size matches the LEOPARDSat-1 heritage specification (16-byte IV + up to 223 bytes combined header/ciphertext + 16-byte ASCON tag = 255 bytes max).	L3-OBC-129, L3-OBC-130, L4-OBC-185, L4-OBC-186	Kristen Griessman	2026-11-04
OQ-01	Open Question	**Secondary Payload Selection & UART Protocol Details (0xB1–0xD1):** Finalize partner agreement (MyRadar candidate) and confirm exact payload data block lengths.	L3-OBC-137, Secondary Payload UART API	Caden O'Leary	2026-11-18
OQ-02	Open Question	**Onboard Data Storage Capacity (L3-OBC-144 TBD GB):** Confirm whether the flight storage board populates 8 GB or 16 GB SPI NAND Flash alongside the SPI MRAM state store.	L3-OBC-144, L4-OBC-226–230	Seunghun Lee	2026-11-11
OQ-03	Open Question	**OBC Standby & Peak Power Thresholds (L3-OBC-157, L3-OBC-158 TBD W) & Mass Limit (L3-OBC-164 TBD g):** Lock final system-allocated power (0.45 W standby / 2.50 W peak) and mass (250 g) budgets with the EPS and Systems teams.	L3-OBC-157, L3-OBC-158, L3-OBC-164	Seunghun Lee & Caden O'Leary	2026-11-11
OQ-04	Open Question	**EPS-to-UHF Radio Emergency Failover Timing (ADR-004):** Confirm exact UART arbitration timeout when the Power Computer takes over UHF beacon transmission during a Main Computer fault.	L3-OBC-161, L4-OBC-258	Matthew Goldsberry	2026-11-04